

EFPB Research Pipeline

Experiment run report: calibration, freeze and readiness

Runs of 19 September 2026 on one laptop (AMD Ryzen 7 5800U, 8 cores / 16 threads, 30 GB RAM, Ubuntu Linux)

1. Summary

- **The experiment runs on one computer** (8 parallel workers); the two-computer split in the instructions only divides the work. Both backends work end to end: the surrogate and the real SUMO/TraCI runner with virtual pedestrians.
- **Ten defects were found and fixed** on the way (section 3), among them a wrong SUMO signal program that made every SUMO controller drive the wrong phases, missing yellow/all-red clearance, a parallel-run port collision, a surrogate that served more than its configured capacity, and waits that ignored people still queued at the end (which made starving a mode look good). One of them (the split-scaling bug, defect 9) was introduced by me during calibration and caught before it was used.
- **Calibration is done and frozen for both backends** (sections 4 and 5): two demand levels near and above saturation were added, every controller was tuned under the same budget (30 configurations), the picks were confirmed on fresh seeds, and the frozen parameters and a preregistration draft are in the repository. In the surrogate only efpb changes from its defaults; in SUMO every controller keeps its defaults.
- **You are not yet ready to run the confirmatory experiments.** Open blockers: the ablation variants are not implemented, equal_bargaining collapses under saturation and needs a decision, the SUMO runner cannot run the robustness scenarios, pedestrians are virtual, the preregistration is only a draft until you register it, and the HCI study and the statistical analysis do not exist yet (section 8).

2. Running on one computer

Each process runs one job at a time, so the speed-up comes from starting several processes, each taking a different shard. Measured on this laptop with 8 workers:

Run	Runs	Wall time
Surrogate calibration selection (80 configurations, lean mode)	3,200	about 23 s
Surrogate confirmation and delta sensitivity (fresh seeds)	2,400	about 22 s
SUMO calibration selection (80 configurations, all controllers)	2,800	34 min 36 s
SUMO confirmation and delta sensitivity (fresh seeds)	700	8 min 42 s
SUMO pilot (4 controllers x 3 scenarios x 3 seeds)	36	about 50 s
Old 1,540-run surrogate matrix (earlier simulator versions, now stale)	1,540	58 to 73 min

The instructions' estimate of 6.4–8.6 hours on two computers assumed 20 s per run; per-run cost differs by about 30 times between controllers. A full-length SUMO matrix has not been timed. Lean mode (no traces or per-second files) makes surrogate runs about 500 times faster and gives identical metrics.

3. Defects found and fixed

#	Defect	Effect	Fix
1	SUMO signal program: <code>cross.tls.add.xml</code> defines a second program that SUMO activates; the runner's phase numbers belonged to the network's own program 0.	In the first pilot V_NS was almost all red and V_EW fully red; every controller drove the wrong phases. equal_bargaining completed 0 vehicles.	Program 0 is selected; approach pairs 1+2 and 3+4 match the signal.
2	SUMO ignored vehicle demand; the controller was told the vehicle arrival rate was 0.	The 'vehicle heavy' scenario equalled the balanced one.	Bundled flows are rescaled to the configured rate; the rate is passed to the controller.
3	SUMO metrics: wait counted time since first seen, warm-up was included, phase switches counted re-issued phases, pedestrians served counted phase steps.	P95 waits above the run length; identical switch counts for every controller.	Waits use consecutive stopped time; metrics cover the evaluation window; real counts.
4	No yellow / all-red clearance between phases (SUMO and surrogate).	79 emergency stops in the 36 SUMO pilot runs. The surrogate charged no lost time for switching.	yellow 3 s and all-red 1 s in both backends: 0 emergency stops, 0 collisions.
5	Surrogate metrics covered the warm-up; the surrogate controller received arrival rates of 0.	Throughput about 16% too high.	Metrics follow arrivals after warm-up; nominal rates are passed (surrogate 0.2.0).
6	Parallel SUMO runs could pick the same TraCI port; one client could read the other run's simulation.	1 of 36 regenerated runs differed; found by a bind error in its log.	Serialised or per-worker ports plus a check of seed, scale and log path after connecting.
7	Surrogate capacity was rounded up (0.8 to 1 vehicle per second per approach), and the demand generator capped arrivals at 0.95 per second.	The surrogate served 25% more than configured and could not produce saturating demand.	Fractional service credit; arrival cap lifted; existing demand streams unchanged (surrogate 0.3.0).
8	Waits ignored people still queued at the end of a run.	A controller that starved a mode looked good on max wait (equal_bargaining showed a pedestrian wait of 0 with 0% served).	Unserved people count with the wait so far (surrogate 0.4.0, SUMO runner 0.5.0).
9	Static fixed-time splits were scaled by the config's cycle (37) instead of their own total (90).	In the new sweep one vehicle phase was never served, so a broken plan won the first selection.	Scale by the splits' own total and refuse a phase with no time. Caught before use; the first selection was discarded.
10	The default fixed-time baseline (static, 90 s) was untuned.	The research plan forbids comparing a tuned proposal with an untuned default.	Demand-based plan added and calibrated; a 37 s cycle wins in both backends (section 5).

Still open (not defects fixed): the instructions' two-computer commands skip the ablations, their runtime estimate is out of date, the pedestrian clearance interval is not modelled, and the SUMO runner ignores the emergency, sensor-error, accessibility and bursty scenario fields.

4. Demand levels closer to saturation (step 2)

A ladder of demand levels (multiples of the previous top level X: 0.15 pedestrians and 0.44 vehicles per second) was run with every controller at its then-current defaults, on calibration seeds only. Cells show the worst service ratio (share of arrivals served) and the largest wait divided by its 100 s limit (people still queued count with the wait so far).

Surrogate (worst service ratio / max wait per limit):

x X	ped / veh per s	fixed_time	efficiency	equal_bargaining	efpb
0.5	0.075 / 0.22	0.99 / 0.29	0.98 / 0.54	0.99 / 0.56	0.97 / 0.48
1.0	0.150 / 0.44	0.98 / 0.38	0.98 / 0.53	0.98 / 0.53	0.98 / 0.53
1.5	0.225 / 0.66	0.98 / 0.53	0.97 / 0.57	0.96 / 1.30	0.97 / 0.55
2.0	0.300 / 0.88	0.87 / 1.53	0.92 / 1.80	0.83 / 3.54	0.66 / 3.33
2.5	0.375 / 1.10	0.66 / 2.96	0.63 / 6.43	0.00 / 8.99	0.62 / 6.49
3.0	0.450 / 1.32	0.56 / 4.07	0.47 / 8.99	0.00 / 8.98	0.56 / 7.51

x X	ped / veh per s	fixed_time	efficiency	equal_bargaining	efpb
3.5	0.525 / 1.54	0.46 / 5.08	0.35 / 8.99	0.00 / 9.00	0.53 / 7.49
4.0	0.600 / 1.76	0.39 / 5.65	0.25 / 8.99	0.00 / 8.99	0.32 / 8.69

SUMO (share of vehicle demand completed per hour / mean vehicles queued):

x X	veh per s	fixed_time	efficiency	efpb
1.0	0.44	1.01 / 3.8	1.00 / 3.3	1.00 / 3.4
1.5	0.66	0.88 / 19.7	0.91 / 18.5	0.90 / 19.2
2.0	0.88	0.69 / 29.4	0.70 / 30.8	0.67 / 36.7
2.5	1.10	0.58 / 39.2	0.52 / 44.3	0.49 / 46.2
3.0	1.32	0.48 / 39.8	0.26 / 57.7	0.25 / 57.6

- The old top level X is about half of capacity, which is why the controllers barely differed before. The surrogate saturates between 1.5x and 2x X; SUMO earlier, its vehicle throughput plateauing near 2,200 vehicles per hour (about 0.62 per second).
- **New levels** (in configs/design_base.yaml, existing levels unchanged): **S** = 1.5x X (0.225 pedestrians, 0.66 vehicles per second), near saturation, and **O** = 2.0x X (0.30 and 0.88), over saturation. Existing configs and the fingerprints of stored runs are untouched.
- A stress experiment (configs/experiments/stress.yaml: cells S/S, O/O, H/S, S/H, 30 seeds) was added for the confirmatory phase.

5. Calibration and freeze (steps 1 and 3)

5.1 Policy, decided before any confirmation run

- **Equal tuning budget.** Every controller gets at most 30 configurations from its own search space, defaults always included; spaces of 30 or fewer are searched exhaustively, larger ones by seeded random search. fixed_time: plan family and cycle (14). efficiency: horizon, service duration, switch cost (27). equal_bargaining: horizon, service duration (9; its equal weights are by definition not tuned). efpb: horizon, service duration, delta, kappa, eta (30 of 180). Safety timings (minimum green, yellow, all-red) are not tuned.
- **Objectives** over everyone who arrived after warm-up: person delay (occupancy-weighted mean wait) and max normalised wait (largest wait / 100 s). Scenarios weighted equally: surrogate MM, LH, HL, HH, XX, SS, OO and bursty HH; SUMO the same without the bursty cell. Selection seeds 301–305.
- **Selection rule:** per controller, scale both objectives to 0–1 within that controller and take the configuration closest to the ideal point; an exact tie keeps the default; in SUMO a configuration with a collision is excluded. The picks are written to selection.json **before** any confirmation run and cannot be re-picked.
- **Adoption rule** on fresh seeds (surrogate 321–350, SUMO 361–370): a pick replaces the default unless it is worse in both objectives (paired, identical scenarios and seeds). Delta is also run over the plan's set {0, .02, .05, .10, .20}.

5.2 surrogate results

Selection stage (seeds 301–305):

Controller	Configs	Delay range (s)	Max wait range	Default (delay / max wait)	Pick	Adopted?
fixed_time	14	18.6–38.2	0.50–1.46	18.6 / 0.50	default	yes
efficiency	27	17.1–17.8	0.70–0.86	17.1 / 0.73	default	yes
equal_bargaining	9	21.2–23.7	0.99–1.55	21.2 / 0.99	default	yes

Controller	Configs	Delay range (s)	Max wait range	Default (delay / max wait)	Pick	Adopted?
efpb	30	17.6–23.2	0.70–1.27	18.2 / 0.87	eta 2, kappa 4, delta 0.05, horizon 60, duration 15	yes

Confirmation on fresh seeds (defaults, picks and efpb's delta set):

Configuration confirmed on fresh seeds	Controller	Delay (s)	Max wait	Worst scenario
horizon 60, switch cost 5, duration 25 (default)	efficiency	16.4	0.68	1.43
eta 1.0, kappa 2.0, delta 0.05, horizon 60, duration 25 (default)	efpb	17.2	0.73	2.21
eta 2, kappa 4, delta 0.05, horizon 60, duration 15	efpb	17.4	0.69	1.92
efpb delta 0	efpb	17.4	0.69	1.88
efpb delta 0.02	efpb	17.4	0.69	1.88
efpb delta 0.1	efpb	17.3	0.69	1.91
efpb delta 0.2	efpb	17.3	0.69	1.93
horizon 60, duration 25 (default)	equal_bargaining	19.5	0.99	3.59
cycle 37, plan demand_based (default)	fixed_time	18.2	0.51	1.52

Frozen (configs/frozen_surrogate.yaml): {"signal": {"fixed_cycle_s": 37, "fixed_timing": "demand_based"}, "per_controller": {"efpb": {"controller": {"debt_eta": 2, "debt_kappa": 4}, "signal": {"service_duration_s": 15}}}}

delta	Person delay (s)	Max wait	Max wait vs picked
0	17.38	0.6865	-0.0050 ± 0.0045
0.02	17.38	0.6865	-0.0050 ± 0.0045
0.05	17.37	0.6915	+0.0000 ± 0.0000
0.1	17.33	0.6874	-0.0041 ± 0.0022
0.2	17.34	0.6863	-0.0053 ± 0.0028

5.3 SUMO results

Selection stage (seeds 301–305):

Controller	Configs	Delay range (s)	Max wait range	Default (delay / max wait)	Pick	Adopted?
fixed_time	14	20.4–46.7	0.50–1.75	20.4 / 0.50	default	yes
efficiency	27	16.9–17.4	0.49–0.53	16.9 / 0.50	horizon 30, switch cost 0, duration 25	no (kept default)
equal_bargaining	9	30.3–44.4	2.16–2.42	30.3 / 2.16	default	yes
efpb	30	16.0–39.1	0.35–1.36	16.8 / 0.42	eta 2, kappa 2, delta 0.05, horizon 90, duration 35	no (kept default)

Confirmation on fresh seeds (defaults, picks and efpb's delta set):

Configuration confirmed on fresh seeds	Controller	Delay (s)	Max wait	Worst scenario
horizon 30, switch cost 0, duration 25	efficiency	16.7	0.51	0.86
horizon 60, switch cost 5, duration 25 (default)	efficiency	16.6	0.51	0.88
eta 1.0, kappa 2.0, delta 0.05, horizon 60, duration 25 (default)	efpb	17.3	0.44	0.58
eta 2, kappa 2, delta 0.05, horizon 90, duration 35	efpb	19.3	0.48	1.46
efpb delta 0	efpb	19.3	0.48	1.46
efpb delta 0.02	efpb	19.3	0.48	1.46

Configuration confirmed on fresh seeds	Controller	Delay (s)	Max wait	Worst scenario
efpb delta 0.1	efpb	19.3	0.48	1.46
efpb delta 0.2	efpb	19.3	0.48	1.46
horizon 60, duration 25 (default)	equal_bargaining	30.6	2.23	8.99
cycle 37, plan demand_based (default)	fixed_time	24.1	0.62	2.52

Frozen (configs/frozen_sumo.yaml): {"signal": {"fixed_cycle_s": 37, "fixed_timing": "demand_based"}}

delta	Person delay (s)	Max wait	Max wait vs picked
0	19.29	0.4846	+0.0000 ± 0.0000
0.02	19.29	0.4846	+0.0000 ± 0.0000
0.05	19.29	0.4846	+0.0000 ± 0.0000
0.1	19.29	0.4830	-0.0016 ± 0.0015
0.2	19.33	0.4826	-0.0020 ± 0.0023

5.4 The fixed-time cycle under stress

Cycle (s)	Surrogate: delay / max wait / worst scenario	SUMO: delay / max wait / worst scenario
37	18.6 / 0.50 / 1.53	20.4 / 0.50 / 1.59
40	20.9 / 0.60 / 1.92	21.3 / 0.64 / 2.27
45	21.7 / 0.62 / 1.95	20.8 / 0.58 / 1.81
50	19.8 / 0.54 / 1.30	20.8 / 0.53 / 1.34
60	22.5 / 0.62 / 1.44	24.1 / 0.58 / 1.28
75	24.6 / 0.67 / 1.26	27.5 / 0.63 / 1.12
90	27.1 / 0.74 / 1.21	31.4 / 0.72 / 1.09
120	34.3 / 0.95 / 1.36	39.3 / 0.94 / 1.15

Demand-based plans (splits from each scenario's nominal demand). Both backends pick 37 s, the shortest cycle that fits three 10 s minimum greens plus clearance. Longer cycles lower the worst-scenario wait (the over-saturated cell) but raise the overall delay, so the equal-weight rule keeps the short cycle. The static plans with proportional splits had a much higher max wait at every cycle length.

5.5 What the calibration shows, and what to be careful about

- **No controller is clearly ahead.** On the frozen configurations (fresh seeds): surrogate person delay / max wait – fixed_time 18.2 / 0.51; efficiency 16.4 / 0.68; equal_bargaining 19.5 / 0.99; efpb 17.4 / 0.69. SUMO – fixed_time 24.1 / 0.62; efficiency 16.6 / 0.51; equal_bargaining 30.6 / 2.23; efpb 17.3 / 0.44. The ranking differs by backend and by objective. This is calibration output, not a finding.
- **Delta barely matters.** Delta 0 and 0.02 give identical results in both backends (0.05 too in SUMO; in the surrogate it differs by under one percent), and 0.10 and 0.20 change the max wait by less than one percent. The default 0.05 is frozen; the full sensitivity set is reported above, as the plan requires.
- **equal_bargaining collapses under saturation** and none of its tuning parameters helps: at the S and O levels pedestrians wait almost the whole run (max wait about 4 and 9 times the limit in SUMO). The cause is its individual-rationality rule, which never leaves a vehicle phase while vehicles are queued. Decide whether this is the intended baseline before the confirmatory runs; if it is not, comparisons with it will exaggerate any advantage of efpb.
- **The two backends freeze different parameters:** in the surrogate efpb changes (kappa 4, eta 2, service duration 15 s; a small gain in max wait for a small loss in delay), in SUMO every controller keeps its defaults because the picks did not survive the fresh seeds. The SUMO frozen set is the one to use for SUMO experiments; the surrogate is engineering validation only.
- **Equal weights make the over-saturated cell dominate** the averages. Per-scenario numbers are in the result files; a different weighting would be a deviation to preregister, not a tweak.

Max normalised wait by scenario for each controller's defaults (selection seeds; above 1 the limit is exceeded):

surrogate	K_MM	K_LH	K_HL	K_HH	K_XX	K_SS	K_OO	K_HH_burst
fixed_time	0.29	0.30	0.29	0.29	0.38	0.53	1.53	0.42
efficiency	0.54	0.86	0.55	0.49	0.53	0.57	1.80	0.50
equal_bargaining	0.47	0.47	0.49	0.50	0.53	1.30	3.54	0.62
efpb	0.51	0.54	0.44	0.51	0.53	0.55	3.33	0.54

SUMO	K_MM	K_LH	K_HL	K_HH	K_XX	K_SS	K_OO
fixed_time	0.27	0.26	0.27	0.27	0.32	0.53	1.59
efficiency	0.31	0.51	0.30	0.45	0.41	0.71	0.84
equal_bargaining	0.34	0.33	0.27	0.45	0.40	4.37	8.95
efpb	0.30	0.46	0.27	0.43	0.39	0.56	0.56

5.6 Disclosures

- A first surrogate selection was made on flawed data (defect 9, and waits that ignored unserved people). I saw its confirmation results before discarding it; everything shown here was redone from scratch with the corrected code.
- An exact tie in the selection rule once changed a parameter for no measurable reason (efficiency's horizon 30 and 60 give identical results). The rule was refined so that ties keep the default; I saw that tie before changing the rule.
- Selection used the same seeds for all 80 configurations; the confirmation on fresh seeds is what guards against the optimism of picking the best of many.
- Nothing here is registered. docs/preregistration_calibration.md is a draft generated from the frozen files; it only becomes a preregistration when you register it with a timestamp outside this repository, before the confirmatory runs.

6. The SUMO pilot (time-boxed, virtual pedestrians)

36 runs: 3 scenarios x 4 controllers x 3 seeds (9001–9003, never used for calibration). fixed_time uses the tuned demand-based 37 s plan. The pilot config keeps the provisional parameters (identical to the SUMO frozen set); all 36 runs completed with no collisions and no emergency stops. Means of 3 seeds.

Scenario	Controller	Veh/h	Stop time (s)	Wait p95 (s)	Queue	Ped served	Ped wait (s)	Switches	Clear. %
S1_balanced	fixed_time	792	7.5	18.7	1.63	65.0	10.4	73	18.7
S1_balanced	efficiency	783	7.3	16.7	1.55	65.3	9.0	72	18.1
S1_balanced	equal_bargaining	788	8.3	20.3	1.76	63.3	6.1	72	17.0
S1_balanced	efpb	780	6.9	17.0	1.47	65.3	8.7	72	18.1
S2_pedestrian_heavy	fixed_time	792	7.5	18.7	1.63	101.3	11.6	73	18.7
S2_pedestrian_heavy	efficiency	789	8.5	17.3	1.81	99.7	8.5	73	17.6
S2_pedestrian_heavy	equal_bargaining	780	9.0	17.7	1.91	99.0	6.7	74	16.7
S2_pedestrian_heavy	efpb	789	8.1	17.0	1.72	99.7	8.1	73	17.7
S3_vehicle_heavy	fixed_time	1239	9.4	19.0	3.02	65.0	10.4	73	18.7
S3_vehicle_heavy	efficiency	1219	7.4	17.7	2.34	64.7	13.2	72	20.0
S3_vehicle_heavy	equal_bargaining	1236	9.0	22.7	2.83	64.7	6.5	73	17.9
S3_vehicle_heavy	efpb	1229	7.4	18.7	2.31	63.3	10.6	71	20.1

Tuned fixed-time minus each adaptive controller (same scenarios and seeds; mean $\pm$ standard error, 9 pairs; negative = fixed_time lower):

Metric	vs efficiency	vs equal_bargaining	vs efpb
Vehicle stop time (s)	+0.39 $\pm$ 0.47	-0.63 $\pm$ 0.33	+0.67 $\pm$ 0.39
Vehicle wait p95 (s)	+1.56 $\pm$ 0.34	-1.44 $\pm$ 0.87	+1.22 $\pm$ 0.55
Pedestrian mean wait (s)	+0.54 $\pm$ 0.93	+4.31 $\pm$ 0.26	+1.65 $\pm$ 0.61
Pedestrian max wait (s)	-9.89 $\pm$ 3.25	+2.78 $\pm$ 0.95	-7.11 $\pm$ 2.46

- **Vehicles:** no controller is clearly ahead (stop times within about 1 s). **Pedestrians:** the trade-offs differ; equal_bargaining has the lowest pedestrian waits at these (light) demand levels. Only 3 seeds, and fixed_time's vehicle results are identical in S1 and S2 (its timing ignores pedestrians), so the standard errors are optimistic. Treat this as descriptive.
- **Effect of clearance** (mean over the pilot; pilot with the old fixed_time plan, static 90 s, so that timing is not mixed in): mean vehicle stop time without clearance / with clearance – fixed_time 16.1 / 16.5 s; efficiency 4.6 / 7.7 s; equal_bargaining 5.5 / 8.8 s; efpb 4.4 / 7.5 s. Emergency stops: 79 without clearance, 0 with.
- Tuning the fixed-time plan (same seeds): mean vehicle stop time 16.5 s with the static 90 s plan against 8.1 s with the tuned plan.

7. Integrity and reproducibility

- **Versions and fingerprints.** Every stored run carries the simulator version (surrogate 0.4.0, SUMO runner 0.5.0) and a fingerprint of everything that decides its result (settings, scenario, controller, seed, plus the SUMO input files). A stored run made with another configuration or version is refused, not reused. Bump the versions by hand when a code change alters results.
- **Pre-committed selection.** freeze_calibration.py writes the picks before any confirmation run and refuses to re-pick if the selection results change.
- **Reproducibility.** Calibration, sweep and pilot outputs reproduced exactly when regenerated (checked before each version change); 49 automated tests pass, including a real-SUMO end-to-end test and tests for most of the defects above.
- **Calibration-only data.** Calibration scenarios and seeds are refused if they overlap the confirmatory, robustness, ablation, stress or SUMO-pilot experiments.
- **Stale results are marked.** data/RESULTS_STATUS.md lists what is current and what is not. The old 1,540-run surrogate matrix predates surrogate 0.3.0/0.4.0, the frozen parameters and the stress levels; it must be moved aside before the matrix is run once.

8. Readiness for the confirmatory experiment

Item	Status	What is missing
Pipeline, reproducibility, safeguards	Ready	–
Demand levels (S, O) and stress experiment	Done	–
Tuning-budget policy, calibration, freeze (both backends)	Done	Register the preregistration draft; decide the weighting of scenarios if you want a different one.
Surrogate matrix (confirmatory_core 1,080, robustness 360, stress 480)	Can run (~1.5 h)	Move the stale data aside first. Engineering validation only.
Ablations (100 runs)	Not ready	The ablated controllers are not implemented; the five cells are identical.
equal_bargaining baseline	Needs decision	Collapses under saturation; confirm it is the intended baseline.
SUMO experiment at matrix scale	Not ready	Runner ignores emergency, sensor, accessibility and bursty scenarios; pedestrians are virtual; no matrix configs or runtime measurement.
HCI participant study	Not started	Dashboard, participant app, ethics approval, participants.
Statistical analysis and paper	Not started	Preregistered models (GLMM, LMM, ordinal) are not implemented; results table is still TBD.

Suggested next steps: (1) register the preregistration; (2) decide on equal_bargaining; (3) implement the ablation variants; (4) decide the SUMO scope (extend the runner to the robustness scenarios, or limit SUMO claims to the nominal and stress cells) and add SUMO matrix configs; (5) run the surrogate matrix once and implement the preregistered analysis; (6) build the participant study in parallel.

9. Research-integrity note

Nothing in this report is a research finding. The surrogate is an approximation, the SUMO runs use virtual pedestrians, calibration output only chooses parameters, and no participant study has been carried out. Results from the first, invalid SUMO pilots and from the flawed first calibration were discarded, not used. All tables are generated from the stored result files by `docs/make_report.py`.

Appendix: where things are

What	Where (inside efpb-research-pipeline/)
Frozen parameters	configs/frozen_surrogate.yaml, configs/frozen_sumo.yaml
Preregistration draft	docs/preregistration_calibration.md (regenerate with <code>make_preregistration.py</code>)
Demand levels, experiment configs	configs/design_base.yaml; configs/experiments/{confirmatory_core, robustness, ablations, stress}.yaml
Surrogate calibration	data/calibration_sweep/ (selection.json, summary.csv, freeze_report.md, confirmation/)
SUMO calibration	data/sumo_calibration/ (selection.json, summary_selection.csv, freeze_report.md, selection/, confirmation/)
Saturation study	data/saturation_study/
SUMO pilot and comparison sets	data/sumo_runs/, data/sumo_runs_static_fixed_time/, data/sumo_runs_v2_no_clearance/, data/sumo_fixed_time_check/
Stale or archived	data/runs/ (stale), data/runs_v1_no_clearance/, data/sumo_runs_v1_defective/, data/archive_*/
Status of every result	data/RESULTS_STATUS.md